6.4 Sequence of Transformations

Lesson Introduction

Benchmarks	MA.912.GR.2.2 Identify transformations that do or do not preserve distance. MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can represent a sequence of transformations that will map the preimage onto the image. • I can describe a sequence of transformations that will map the preimage onto the image. • I can identify transformations that do or do not preserve distance. • I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.	N/A

Essential Questions	• What transformations occurred to map one shape onto another? • What transformations included in a sequence of transformations don't preserve distance? • How can the orientation of the vertex points on a figure and on the plane help determine the transformations that took place?
Lesson Narrative	This lesson continues exploring sequences of transformations that preserve distance. Students will continue to identify and describe sequences of transformations through written explanations and algebraic descriptions.

Component Summary	6.4.1 Warm-Up (~5 min.)	Would You Rather choice between driving speeds *(SE p. 302)*	6.4.3 Guided Practice (10–15 min.)	Error analysis for a sequence of transformations *(SE p. 305)*
	6.4.2 Guided Instruction (10–15 min.)	Describing and analyzing a sequence of transformations *(SE p. 302)*	6.4.4 Your Turn! (10–15 min.)	Describing and writing a sequence of transformations *(SE p. 306)*
	6.4.5 Wrap-Up (~5 min.)	Determining the sequence of transformations *(SE p. 306)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Transformation • Rotation • Reflection • Translation • Dilation	• (Optional) Patty Paper • (Optional) Copies of Preimages and Images	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may compare the rates in the 6.4.1 Warm-Up without writing them in common units.
Students may write the wrong algebraic description for given transformations.
Students may swap the order of the sequence of transformations because they think they are commutative.
Students may write a dilation as an enlargement by using a number greater than one instead of as a reduction by using a number less than one or vice versa.</td><td>Students may provide an algebraic description that matches an incorrect written description. Be sure to pay close attention to each response to determine remediation needs.
Some questions may have more than one answer. Analyze each student’s response for correct sequences.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Stronger and Clearer Each Time
6.4.2 Guided Instruction is an opportunity for students to revise and refine their ideas and their written output. Using a structured pairing strategy provides opportunities to clarify their responses and revise their original written response.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
6.4.3 Guided Practice is a collaboration opportunity for students. Students get to work together to review pre-created fictional student thinking. They identify and correct errors and justify their thinking comparatively, gaining important practice in this Math Thinking and Reasoning Standard.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Provide kinesthetic and visual entry-points by having students first take a copy of the preimage and physically demonstrate what sequence of transformations will result in the final image. Then students can move on to providing a written description and an algebraic description of the sequence of transformations.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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6.4.1 Warm-Up: Would You Rather?

Component Overview: *Students are given an opportunity to select between driving at two different rates. Regardless of which they choose, make sure students justify their choice with mathematics.*

6.4.1 Warm-Up: Would You Rather?

1. Would you rather drive a car at a rate of 40 miles per hour, or drive a car at a rate of 50 feet per second? Remember: There are 5280 feet in a mile.

Explain your choice using mathematics.
Answers will vary. I would rather drive 40 miles per hour. If I were driving 50 feet per second, I would be driving approximately 34 miles per hour, which would be slower.

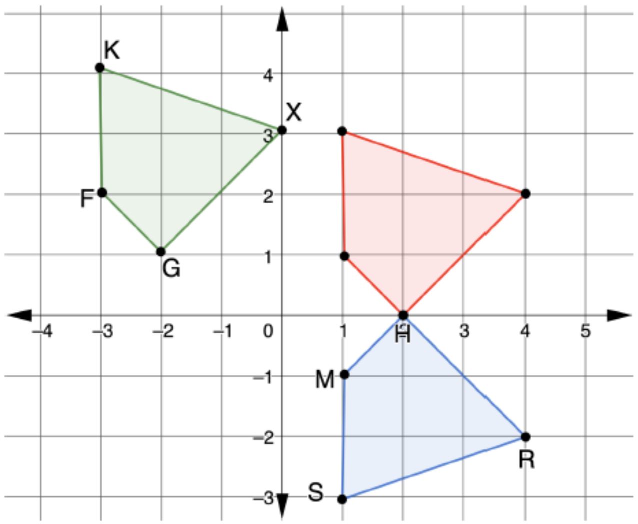
Consider asking students if they would prefer to take a ride traveling 60 kilometers per hour? Sixty kilometers per hour is about 37 miles per hour. This could lead to a discussion about why we have different units of measure in different parts of the world and for different applications.

- *What would happen if you traveled to another country and a speed limit sign showed 70 with no units? What problems could this cause?*

Component Overview: *Students identify and describe three different sequences of transformations to determine if distance was preserved in each. The questions are scaffolded to gradually move away from providing each individual transformation in the sequence. Encourage students to continue to look back at their notes for algebraic descriptions of the different transformations if they cannot fluently recall them.*

6.4.2 Guided Instruction: Sequence of Transformations

1. $XGFK$ is the result of a sequence of transformations of $RSMH$. The red quadrilateral is the first transformation.



   a. Describe the first transformation of $RSMH$.
   A reflection across the x-axis

   b. Write the transformation algebraically.
   $(x, y) \to (x, -y)$

   c. Describe the second transformation of $RSMH$ that results in $XGFK$.
   A translation left 4 units, up 1 unit

   d. Write the transformation algebraically.
   $(x, y) \to (x - 4, y + 1)$

Before moving to questions 1a-1e, have students read the question stem and discuss which figure is the preimage and which are the first and second transformations. This move will assist any students with reading difficulties or ELL students with determining where the sequence starts and where it ends. Students may also benefit from labeling the points in both transformed images with the same letters in the preimage to help determine which transformations occurred in the sequence.

For question 1a and 1b, consider asking the following:

- *What clues in the image helped you determine which transformation occurred? (Possible answer: the vertex points $RSMH$ are clockwise in the preimage but counterclockwise in the image, indicating a reflection occurred.)*
- *What strategy helps you to remember the algebraic description of this transformation?*

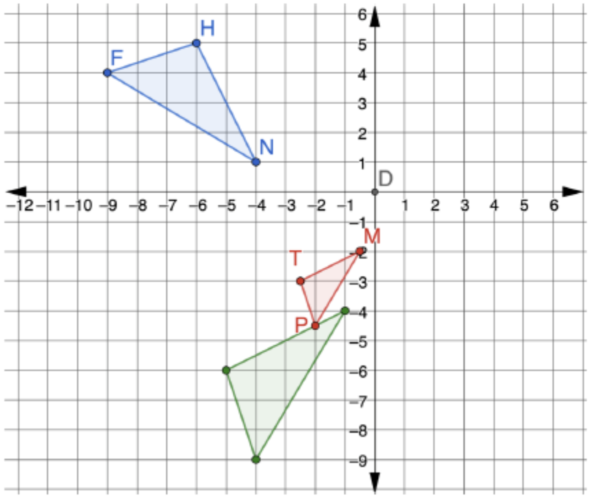
For question 1c and 1d, consider asking the following:

- *What clues in the image helped you determine which transformation occurred? (Possible answer: the orientation of the vertex points $RSMH$ on the image and the orientation of the image on the plane did not change from the previous image, indicating a translation.)*
- *What strategy helps you to remember the algebraic description of this transformation?*

6.4.2 Guided Instruction: Sequence of Transformations (continued)

e. Explain if the sequence of transformations preserves distance.
Both the transformations in the sequence are rigid motions, so they preserve distance.

2. ΔMTP is the result of a sequence of transformations of ΔHNF.



a. Determine the sequence of transformations that will map ΔHNF onto ΔMTP. Complete the table.

	ΔHNF Transformed to $\Delta H'N'F'$	$\Delta H'N'F'$ Transformed to ΔMTP
Written Description	*Rotated 90° counterclockwise about the origin.*	*Dilated through point D with a scale factor of $\frac{1}{2}$*
Algebraic Description	$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$

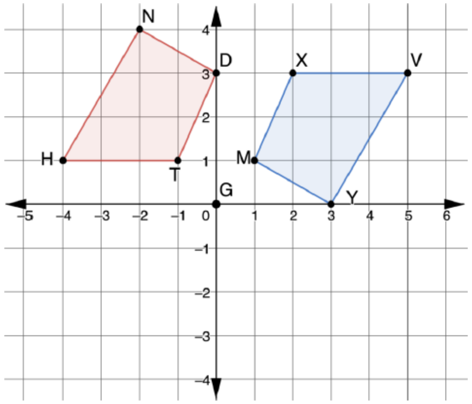
Stronger and Clearer Each Time
For a different strategy, consider putting students in groups of four. Without talking, each student writes the answer to question 1a. When finished, the students pass their answers to the student on their right. The students read the answer for question 1a, write or ask any clarifying questions, and pass it back. The students revise their answer to question 1a and then answer question 1b. The students then pass their answer to the person across from them. This student checks their work, writes or asks clarifying questions, and then passes it back. The students again revise or make any edits and then move to question 1c. This continues until all questions are answered, reviewed by a group member, and then edited/revised if needed.

For question 2a, consider asking the following:

- *What clues in the image helped you determine which transformations occurred? (Possible answer: for the first transformation, the orientation of the vertex points on the image stayed the same, but the orientation of the image on the plane changed from the previous image, indicating a rotation. For the second transformation, the length of the lines change or the image changed size, so I knew it was a dilation.)*
- *What strategy helps you to remember the algebraic description of these transformations?*

b. Explain how to determine if this sequence of transformations preserves distance.
The sequence did not preserve distance. You can tell because there was a dilation that reduced each side by a scale factor of $\frac{1}{2}$. This can also be verified by using the distance formula.

3. $TDNH$ is the result of a sequence of transformations of $XMYV$.



a. Determine a sequence of transformations that will map $XMYV$ onto $TDNH$. Complete the table.
Answers will vary.

	First Transformation	**Second Transformation**
Written Description	*Rotated $180°$ clockwise about point G.*	*Translated up 4 units and to the right 1 unit.*
Algebraic Description	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (x + 1, y + 4)$

b. Which transformations preserve distance?
Both transformations preserve distance. The rotation and translation both maintain side lengths equal to the pre-image.

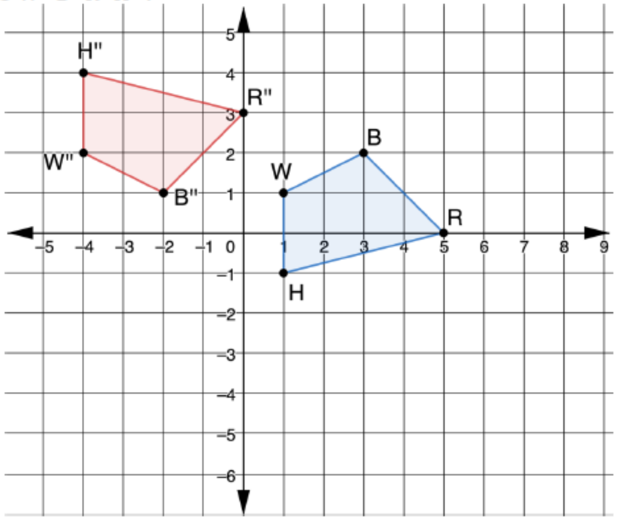
Encourage students to provide a more detailed justification if they respond, "because it is a dilation." Guide the conversation to include the vocabulary "preserved distance" and how using the distance formula or patty paper can verify this.

6.4.3 Guided Practice: Mapping a Given Preimage onto an Image

***Component Overview**: Students consider a transformation and review two sample student works to determine which student determined the correct sequence of transformations that results in an image.*

6.4.3 Guided Practice: Mapping a Given Preimage onto an Image

$WBRH$ was transformed by a sequence of transformations to create $W''B''R''H''$.



Esmeralda and Wesley were determining the sequence of transformations and wrote their answers in the table.

	$WBRH$ **Transformed to** $W'B'R'H'$	$W'B'R'H'$ **Transformed to** $W''B''R''H''$
Esmeralda's Answer	Translated 5 units to the left and 3 units down	Reflected over the x-axis
Wesley's Answer	Translated 3 units down	Rotated $180°$ about the origin

1. Determine whose sequence of transformations is correct. Explain your reasoning.
   Esmeralda is correct because her transformations map $WBRH$ onto $W''B''R''H''$.

Students may use different methods to determine which of the two answers are correct. Some students may use patty paper and a copy of the preimage to manipulate. Others may test the vertex points to determine if they map onto the image. Have students who used different methods share their work with the class. Students benefit from seeing multiple methods and hearing the thinking of others.

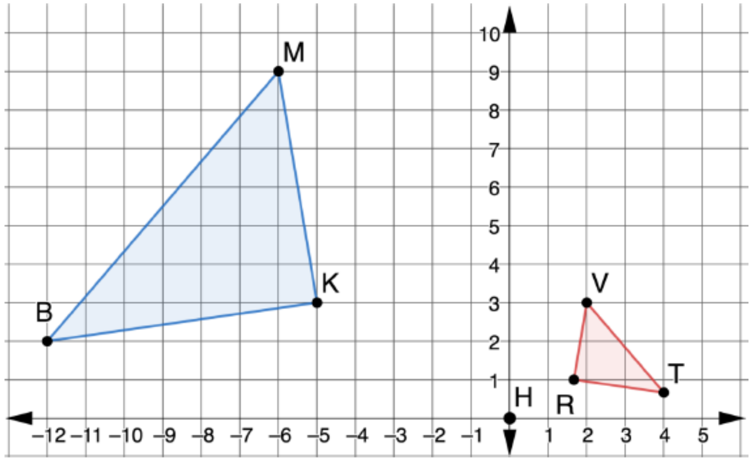
MA.K12.MTR.4.1: Discussion and Reflection
This is an opportunity for students to work collaboratively and explore the mathematical thinking and reasoning of others. Offering for students to work through question 1 with a partner or small group will allow them to explore potential errors and misunderstandings in a safe and effective way.

6.4.4 Your Turn!

Component Overview*: Students describe and write the algebraic description for a sequence of transformations given the preimage and final image.*

6.4.4 Your Turn!

1. ΔMKB was transformed to create ΔVRT.



a. Describe a sequence of transformations that will map ΔMKB onto ΔVRT.
Answers will vary. ΔMKB was reflected across the y-axis and then dilated by a scale factor of $\frac{1}{3}$ from the origin.

b. Write the sequence of transformations using algebraic descriptions.
Answers will vary.
First transformation: $(x, y) \rightarrow (-x, y)$
Second transformation: $(x, y) \rightarrow \left(\frac{1}{3}x, \frac{1}{3}y\right)$

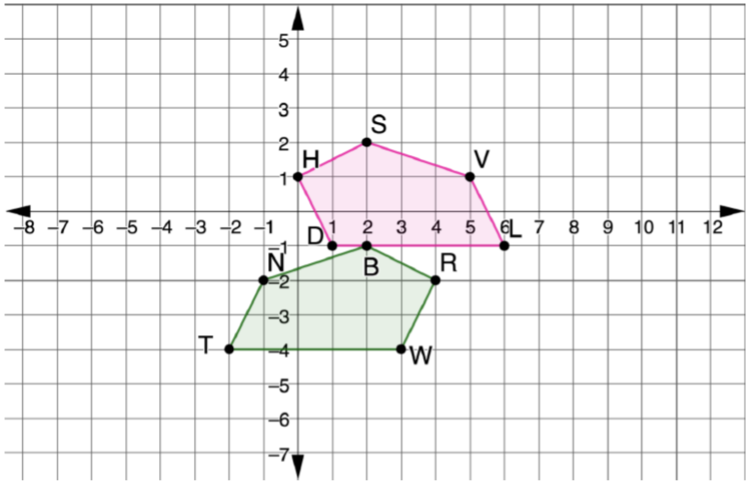
Consider having students record their work on a separate piece of paper without a name. When students are done, mix up the papers and redistribute to the class. The students should comment about the work on the paper. Comments should include both positive (I like the way you…) and constructive (While your first transformation is correct, your second transformation; You are correct in recognizing a dilation, but the scale factor you chose…). Post the results in a Gallery Walk style so anyone can walk by and gain advice without being identified as the one who made a mistake.

6.4.5 Wrap-Up: Ticket Out the Door

***Component Overview**: Students determine the sequence used to map a preimage onto an image in both written form and as an algebraic description.*

6.4.5 Wrap-Up: Ticket Out the Door

1. A sequence of transformations can map $NBRWT$ onto $VSHDL$.



Determine a sequence of transformations that will map $NBRWT$ onto $VSHDL$. Complete the table.
Answers will vary.

	First Transformation	Second Transformation
Written Description	*Reflected over the y-axis*	*Translated up 3 and right 4*
Algebraic Description	$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x + 4, y + 3)$

Use the answers to question 1 to determine remediation steps. If an incorrect written description is given, a student may need more practice with identifying transformations. If an algebraic description doesn't match the written description, students may need strategies (i.e. Foldable with key terms) for remembering the algebraic descriptions of each transformation.

Students may choose a different line of reflection, which will determine the units translated. Be sure to analyze each student's answer as a sequence of moves.